

Experiment 'p_3_10_temperature_altitude' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_10_temperature_altitude

Natural language statement: null

Method signature: p_3_10_temperature_altitude (t: real, a: nat) returns (s: string)

Ensures

- if t < 0.0 then s == "solid" else if t == 0.0 then s == "solid or liquid" else if 0.0 < t < 100.0 - (a as real / 300.0) then s == "liquid" else if t == 100.0 - (a as real / 300.0) then s == "liquid or gaseous" else s == "gaseous"

Requires

- t >= -273.15 && a <= 10000

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_10_temperature_altitude (t: real, a: nat) returns (s: string)`

The method should respect the following contract:

requires `t >= -273.15 && a <= 10000`, ensures if `t < 0.0` then `s == "solid"` else if `t ==`
→ `0.0` then `s == "solid or liquid"` else if `0.0 < t < 100.0 - (a as real / 300.0)` then
→ `s == "liquid"` else if `t == 100.0 - (a as real / 300.0)` then `s == "liquid or`
→ `gaseous"` else `s == "gaseous"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3EBwp5hNvdWuJvgVNU9ckyhNIOe

Model: gpt-5.5-2026-04-23

Created at: 1780582851

Finish reason: stop

Usage: 464 tokens in, and 659 tokens out

```
{
  var boiling: real := 100.0 - (a as real) / 300.0;
  if t < 0.0 {
    s := "solid";
  } else if t == 0.0 {
    s := "solid or liquid";
  } else if 0.0 < t && t < boiling {
    s := "liquid";
  } else if t == boiling {
    s := "liquid or gaseous";
  } else {
    s := "gaseous";
  }
}
```

Final Program

```

method p_3_10_temperature_altitude (t: real, a: nat) returns (s: string)
  requires t >= -273.15 && a <= 10000
  ensures if t < 0.0 then s == "solid" else if t == 0.0 then s == "solid or liquid"
    ↪ else if 0.0 < t < 100.0 - (a as real / 300.0) then s == "liquid" else if t
    ↪ == 100.0 - (a as real / 300.0) then s == "liquid or gaseous" else s == "
    ↪ gaseous"
{
  var boiling: real := 100.0 - (a as real) / 300.0;
  if t < 0.0 {
    s := "solid";
  } else if t == 0.0 {
    s := "solid or liquid";
  } else if 0.0 < t && t < boiling {
    s := "liquid";
  } else if t == boiling {
    s := "liquid or gaseous";
  } else {
    s := "gaseous";
  }
}

```

Total Token Usage

Input tokens: 464

Output tokens: 659

Reasoning tokens: 512

Sum of 'total tokens': 1123

Experiment Timings

Overall Experiment started at 1780582850863, ended at 1780582914126, lasting 63263ms (63.26 seconds)

Iteration #1 started at 1780582850921, ended at 1780582914126, lasting 63205ms (63.21 seconds)